

Option Sets, Agreement, Conflict, Precedence, and Difference

1 Definitions

Define:

1. a **type** τ ;
2. a **key** k , a unique identifier such that $k : \tau$;
3. a **value** v , an optional mapping to a key k that can be anything;
4. an **option** o , whose only requirement is that it has a key, but which may optionally be mapped to a value v ;
5. an **option set** O , containing options $o_1, o_2, \dots, o_n$, if and only if each option's key is unique to this set.

Define the key set of an option set as:

$$K(O) = \{ k(o) \mid o \in O \}.$$

2 Comparing Option Sets

Two option sets are said to be **equal** if and only if they contain the same number of elements and all their elements are identical.

Two option sets are defined as being **in agreement** if and only if, for every key k that occurs in both sets, the corresponding options are identical:

$$O_1 \sim O_2 \iff \forall k \in K(O_1) \cap K(O_2), O_1[k] = O_2[k].$$

Two option sets are defined as being **in conflict** with one another if and only if there exists at least one key that occurs in both sets for which the corresponding options are not identical:

$$O_1 \not\succ O_2 \iff \exists k \in K(O_1) \cap K(O_2) \text{ such that } O_1[k] \neq O_2[k].$$

3 Precedence

$$O_2 \succ O_1$$

means that O_2 has precedence over O_1 .

Define a **precedence-resolved union**:

$$O_1 \oplus O_2.$$

This may be read as:

O_1 is overridden by O_2 .

Equivalently:

Combine O_1 and O_2 , giving precedence to O_2 .

Formally:

$$K(O_1 \oplus O_2) = K(O_1) \cup K(O_2),$$

and:

$$(O_1 \oplus O_2)[k] = \begin{cases} O_2[k], & k \in K(O_2), \\ O_1[k], & k \in K(O_1) \setminus K(O_2). \end{cases}$$

That says:

- keep every option whose key appears in either set;
- if only O_1 has the key, use O_1 's option;
- if O_2 has the key, use O_2 's option;
- therefore, when both sets share a key, O_2 overrides O_1 .

4 Difference with Exceptions

Lastly, define the **exception predicate** ε and write it as:

$$\varepsilon(o).$$

where:

$$\varepsilon(o) = \text{true}$$

means that o is exempt from ordinary difference or conflict comparison.
Then define a general difference function:

$$\text{diff}_\varepsilon(O_a, O_b),$$

where ε controls which options are ignored, exempted, or filtered during comparison.

A simple definition is:

$$\text{diff}_\varepsilon(O_a, O_b) = \{ k \in K(O_a) \cap K(O_b) \mid O_a[k] \neq O_b[k] \wedge \neg \varepsilon(O_a[k]) \wedge \neg \varepsilon(O_b[k]) \}.$$

Meaning:

diff returns the shared keys whose options differ, except where either option is exempted by ε .